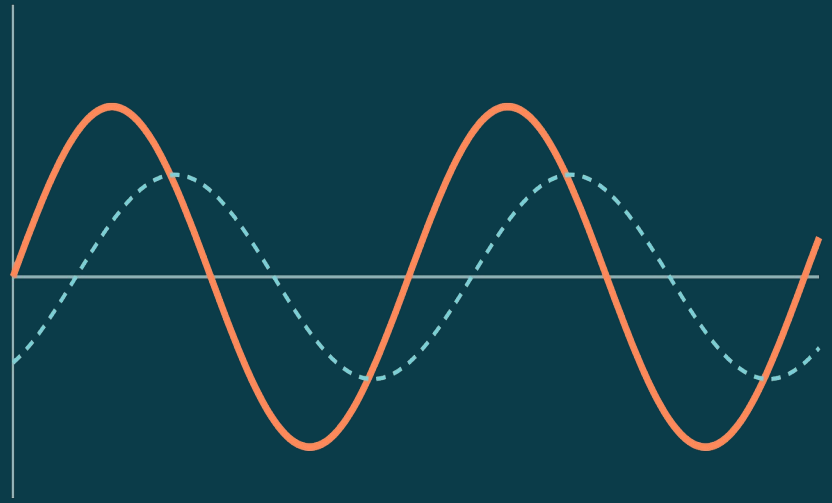
Half-Wave and Bridge Rectifiers



Basic Electrical and Electronics Engineering

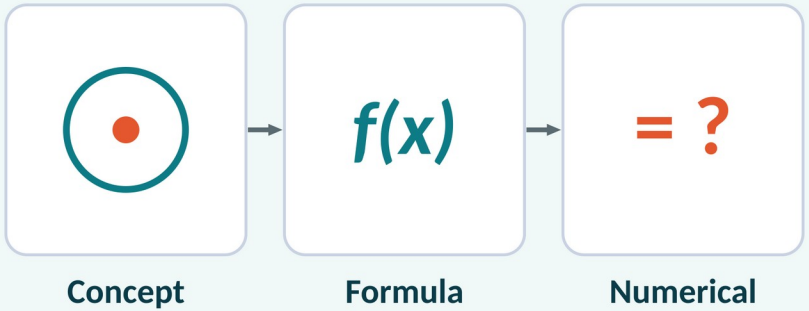
Unit 3: Diodes, Transistors and their Applications

Topic: Half-wave rectifier, full-wave bridge rectifier

Prof. Mitesh Astik

Learning Outcomes

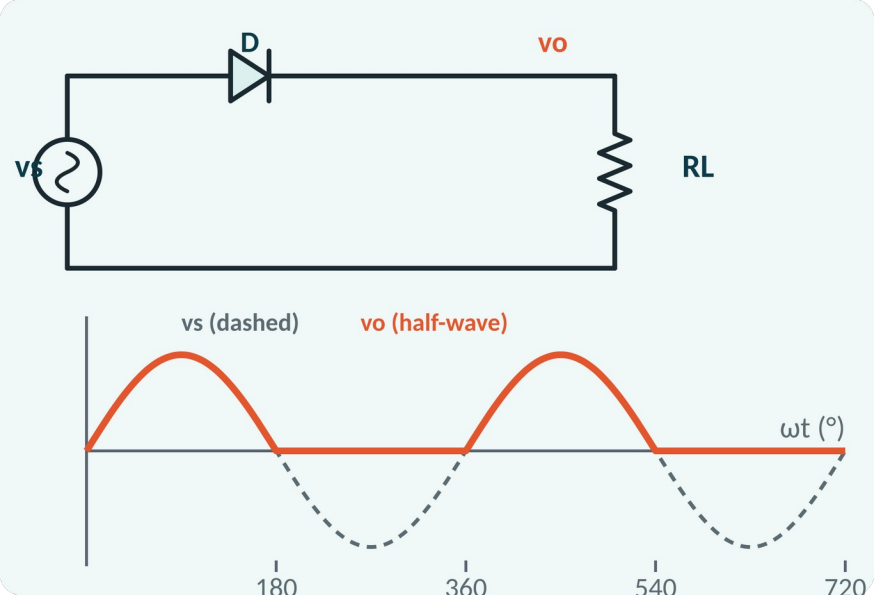
- Draw and explain the half-wave and bridge rectifiers.
- Calculate Vdc, Idc, Vrms, ripple factor, efficiency, PIV.
- Compare the two rectifiers.
- Explain the effect of a capacitor filter.



Linked to CO3
Analyse diode characteristics to understand operation in specific applications.

Recap: What You Already Know

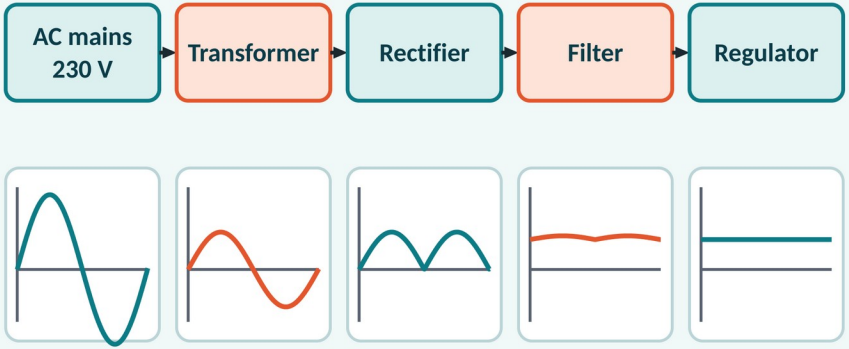
- A diode conducts when forward biased.
- Practical drop $V_\gamma \approx 0.7$ V (Si).
- Sine wave: average over a half cycle 0.637 Vm.
- RMS value of a sine: $V_m/\sqrt{2}$.



Average of a half-wave: V_m/π; RMS of a half-wave: $V_m/2$

Motivation: AC Mains to DC Supply

- Electronics need steady DC.
- Mains is 50 Hz AC.
- Chargers and adapters contain rectifiers.
- Rectifier is the key step.



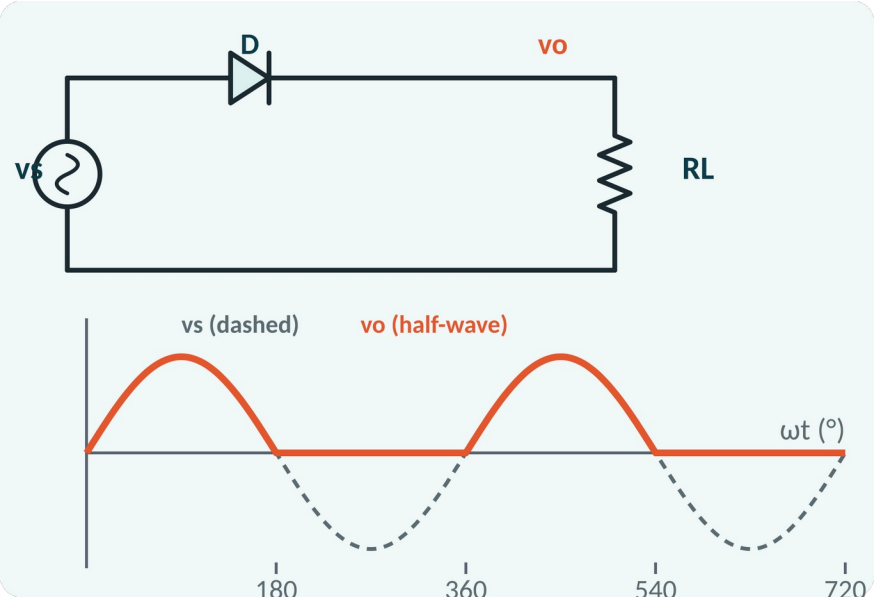
DC supply = transformer + rectifier + filter + regulator

Today: the rectifier stage

AC → transformer → rectifier → filter → regulator → DC load

Half-Wave Rectifier

- One diode in series with the load.
- Positive half-cycle: diode conducts.
- Negative half-cycle: diode blocks.
- Output is a series of positive half-sine pulses.
- Output has a DC component plus ripple.



$v_o = V_m \sin \omega t$ for $0 \le \omega t \le \pi$; $v_o = 0$ for $\pi \le \omega t \le 2\pi$

Half-Wave Rectifier: Performance

- Average value: Vdc = Vm/π.
- RMS value: Vrms = Vm/2.
- Ripple factor 1.21: very poor.
- Efficiency only 40.6 %.
- Diode must withstand Vm in reverse (PIV).

Vdc = Vm/π = 0.318 Vm

Vrms = Vm/2 = 0.5 Vm

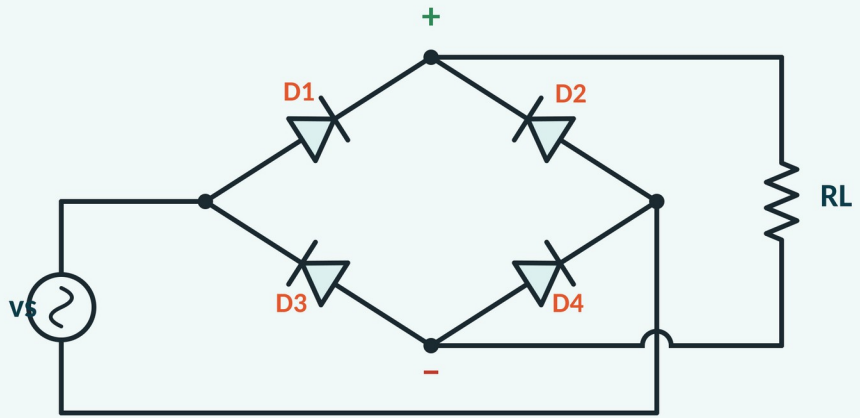
Ripple factor r = 1.21 η = 40.6 %

PIV = Vm form factor = 1.57

Vdc = Vm/π; Idc = Im/π; Vrms = Vm/2; r = √[(Vrms/Vdc)² − 1] = 1.21; PIV = Vm
η = 0.406 (maximum)

Full-Wave Bridge Rectifier

- Four diodes in a bridge.
- Positive half-cycle: D1 and D4 conduct.
- Negative half-cycle: D2 and D3 conduct.
- Current through the load has the same direction.
- No centre-tapped transformer needed.



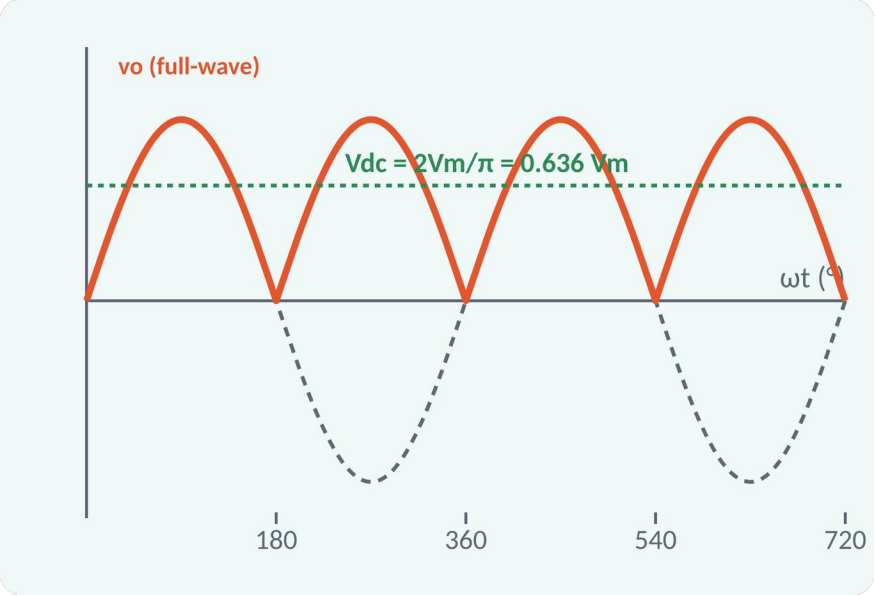
+ half-cycle: D1, D4 conduct – half-cycle: D2, D3 conduct

current through RL always flows the same way

$v_o = |V_m \sin \omega t|$ (minus two diode drops in practice)

Bridge Rectifier: Performance

- Both half-cycles are used.
- Vdc doubles: 2Vm/π.
- Ripple frequency doubles to 2f.
- Efficiency 81.2 %.
- PIV per diode is Vm.



Vdc = 2Vm/π = 0.636Vm; Vrms = Vm/√2; r = 0.482; η = 81.2 %; PIV = Vm

Half-Wave vs Bridge

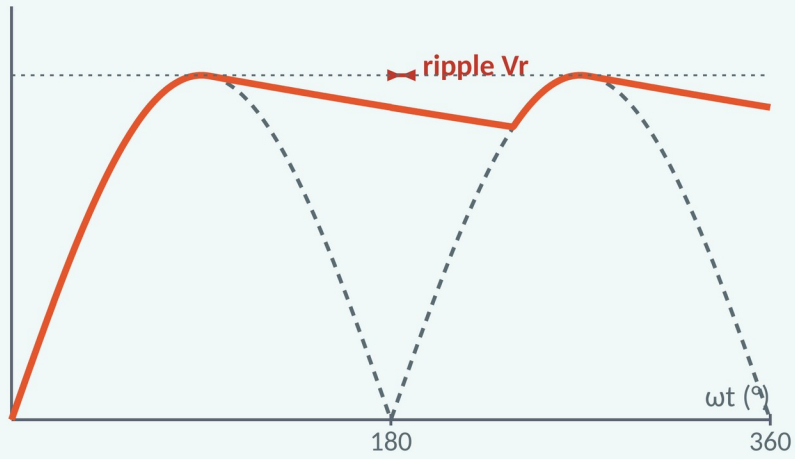
- Bridge: higher Vdc, lower ripple.
- Bridge: twice the efficiency.
- Half-wave: only one diode, simplest.
- Bridge diodes: lower PIV than centre-tap.
- Ripple frequency: f against 2f.

	Half-wave	Bridge
Diodes	1	4
Vdc	0.318 Vm	0.636 Vm
Ripple factor	1.21	0.482
Efficiency	40.6 %	81.2 %
Ripple frequency	f	2f
PIV	Vm	Vm

Prefer the bridge for power supplies; half-wave for very small loads

Capacitor Filter

- Capacitor charges to the peak.
- It discharges slowly through the load.
- Output becomes nearly steady.
- Larger C or RL: smaller ripple.
- Diodes conduct in short pulses.

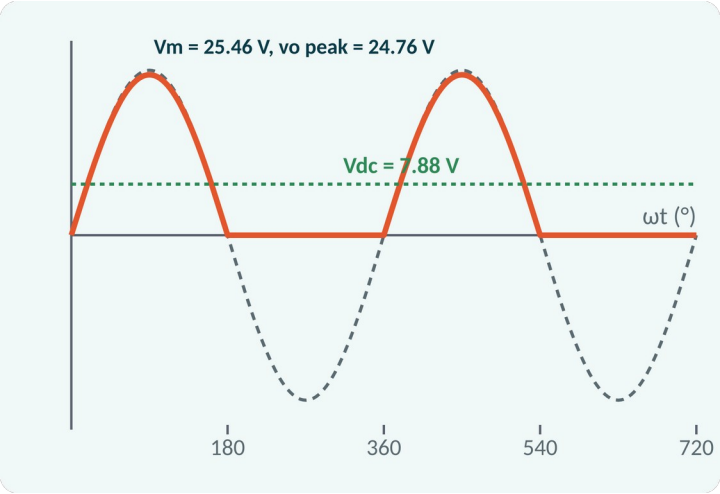


Vdc ≈ Vm − Vr/2; r = 1/(4√3·f·RL·C) for full-wave

Vdc ≈ Vm − Vr/2; r = 1/(4√3·f·RL·C) (full-wave); r = 1/(2√3·f·RL·C) (half-wave)

Solved Example 1: Half-Wave

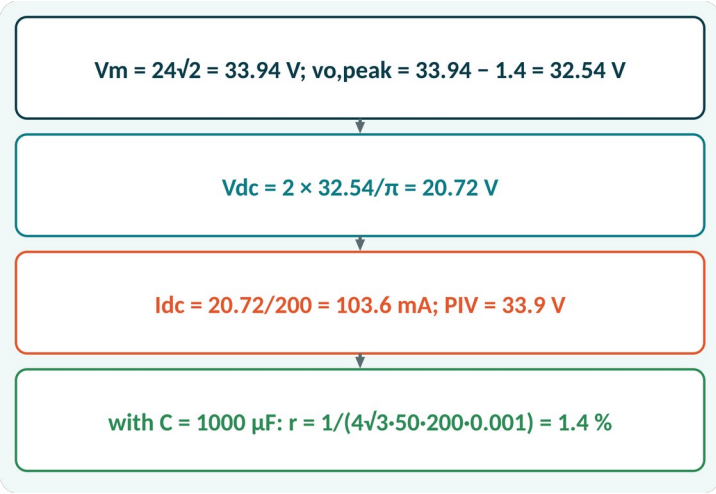
- A half-wave rectifier has a transformer secondary of 18 V RMS, RL = 100 Ω, Si diode. Find Vm, Vdc, Idc, Vrms and PIV.
- 1. Vm = 18√2 = 25.46 V.
- 2. Load peak = 25.46 − 0.7 = 24.76 V.
- 3. Vdc = 24.76/π = 7.88 V; Idc = 78.8 mA.
- 4. Vrms = 24.76/2 = 12.38 V.
- 5. PIV = Vm = 25.46 V.



Answer: Vdc = 7.88 V, Idc = 78.8 mA, Vrms = 12.4 V, PIV = 25.5 V

Solved Example 2: Bridge Rectifier

- A bridge rectifier is fed from 24 V RMS and has RL = 200 Ω, Si diodes. Find Vdc, Idc, PIV and the ripple factor with a 1000 µF capacitor.
- 1. Vm = 24√2 = 33.94 V; load peak = 33.94 − 1.4 = 32.54 V.
- 2. Vdc = 2 × 32.54/π = 20.72 V (without C).
- 3. Idc = 20.72/200 = 103.6 mA.
- 4. PIV = Vm ≈ 33.9 V.
- 5. r = 1/(4√3·50·200·0.001) = 0.0144 (1.4 %).



Answer: Vdc = 20.7 V, Idc = 104 mA, PIV = 33.9 V; r = 1.4 % with 1000 µF

Common Mistakes and Exam Tips

Common mistakes

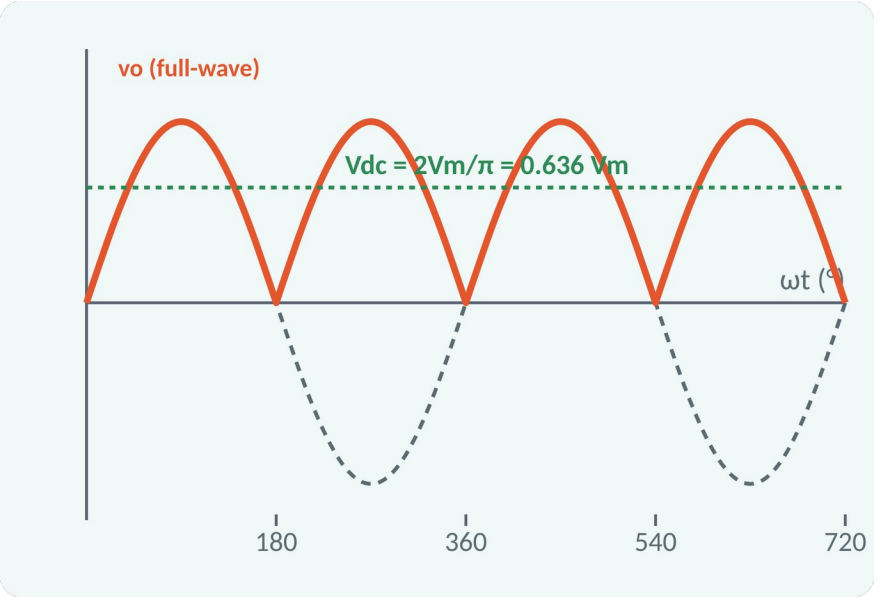
- Using V_m/π for a bridge output.
- Ignoring diode drops in low-voltage circuits.
- Using RMS instead of peak in PIV.
- Forgetting that the bridge conducts two diodes in series.
- Taking ripple frequency as f for a bridge.

Exam tips

- Convert RMS to peak first.
- Subtract 0.7 V (or 1.4 V for a bridge).
- PIV = V_m for the bridge and half-wave.
- State Vdc formulas clearly.

Summary

- Half-wave: one diode, Vdc = Vm/π, ripple 1.21, η = 40.6 %.
- Bridge: four diodes, Vdc = 2Vm/π, ripple 0.48, η = 81.2 %.
- PIV = Vm for both.
- Bridge ripple frequency = 2f.
- Capacitor filter lowers ripple.



Vdc(HW) = Vm/π; Vdc(FW) = 2Vm/π; η = Pdc/Pac

Quick Quiz

Q1. Vdc of a half-wave rectifier with Vm = 10 V:
(a) 3.18 V (b) 6.37 V (c) 5 V (d) 7.07 V

Ans: (a)

Q2. Ripple factor of a full-wave rectifier:
(a) 1.21 (b) 0.48 (c) 0.1 (d) 2

Ans: (b)

Q3. Diodes conducting in a bridge at a time:
(a) 1 (b) 2 (c) 3 (d) 4

Ans: (b)

Q4. Ripple frequency of a bridge on 50 Hz:
(a) 25 Hz (b) 50 Hz (c) 100 Hz (d) 150 Hz

Ans: (c)

Q5. Maximum efficiency of a half-wave rectifier:
(a) 81.2 % (b) 40.6 % (c) 100 % (d) 20 %

Ans: (b)

References

1. **Boylestad Robert L. and Nashelsky Louis, Electronic Devices and Circuit Theory**
   11th Ed., PHI, 2016 (Units 3, 4, 5 textbook)

2. **H. Cotton, Electrical Technology**
   7th Ed., CBS Publishers, 2005 (reference)

Next lecture: Unit 4: BJT construction, operation and use as a switch